

SL Math School: Random Growth Models, etc.

The Brownian Gibbs Property

1 Prologue

1.1 Non-Intersection Events and Acceptance Probabilities

Fix a compact interval $I \subset \mathbb{R}$, $k \in \mathbb{Z}_+$ and vectors $\bar{x}, \bar{y} \in \mathbb{R}_{\geq}^k$, where

$$\mathbb{R}_{\geq}^k = \{\bar{z} \in \mathbb{R}^k : z_1 \geq z_2 \geq \dots \geq z_k\}.$$

Let $\mathcal{B}_{k;\bar{x},\bar{y}}^I$ be the law of k independent Brownian bridges $B_1, \dots, B_k$ on the interval I , so that each B_i starts at x_i and ends at y_i .

For functions $f, g : I \rightarrow \mathbb{R} \cup \{\pm\infty\}$, define the event

$$\text{NoTouch}_{k,I}^{f,g} = \{f(r) > B_1(r) > \dots > B_k(r) > g(r) \text{ for all } r \in I\}.$$

Define also the acceptance probability to be the probability of $\text{NoTouch}_{k,I}^{f,g}$ under the measure $\mathcal{B}_{k;\bar{x},\bar{y}}^I$. That is,

$$A_{k,I}(\bar{x}, \bar{y}; f, g) = \mathcal{B}_{k;\bar{x},\bar{y}}^I(\text{NoTouch}_{k,I}^{f,g}).$$

For shorthand, we often also use the following notation for the law of $\mathcal{B}$ conditioned on NoTouch events:

$$\begin{aligned} \mathbb{P}_{\text{NI}}^{I,\bar{x},\bar{y},f,g} &:= \mathcal{B}_{k;\bar{x},\bar{y}}^I(\cdot \mid \text{NoTouch}_{k,I}^{f,g}), \\ \mathbb{P}_{\text{free}}^{I,\bar{x},\bar{y}} &:= \mathcal{B}_{k;\bar{x},\bar{y}}^I. \end{aligned}$$

Note that if $x_i = x_{i+1}$ or $y_i = y_{i+1}$ for some i , then this conditioning is singular. We can interpret this conditioning as a limit by taking $\tilde{x}_i = x_i - \epsilon$ and taking $\epsilon \rightarrow 0$. Don't worry too much about this.

1.2 A Note On Conditional Expectation and Distributions

For a σ -algebra $\mathcal{F}$ and a random variable X , we define the conditional expectation $\mathbb{E}[X \mid \mathcal{F}]$ to be a $\mathcal{F}$ -measurable random variable which satisfies

$$\mathbb{E}[\mathbb{E}[X \mid \mathcal{F}] \mathbb{1}_C] = \mathbb{E}[X \mathbb{1}_C], \quad \forall C \in \mathcal{F}.$$

Furthermore, we may define the conditional probability $\mathcal{P}_{\mathcal{F}}$ to be so that

$$\mathcal{P}_{\mathcal{F}}(A) := \mathbb{E}[\mathbb{1}_A \mid \mathcal{F}].$$

You may assume that the conditional expectation/distribution is uniquely defined, up to a measure 0 set.

1.3 The Brownian Gibbs Property

For an interval $J \subset \mathbb{N}$, let $\mathcal{L} = \{\mathcal{L}_i\}_{i \in J}$ be a random ensemble of continuous curves, each defined on a common domain $D \subset \mathbb{R}$.

Fix $k_1 \leq k_2$ in J and a compact interval $I = [a, b] \subset D$. We call

$$R = \{k_1, k_1 + 1, \dots, k_2\} \times I$$

a rectangle. The exterior σ -algebra associated to R is

$$\mathcal{F}_{\text{ext}}^{(R)} := \sigma \{ \mathcal{L}_i(x) : (i, x) \notin \{k_1, k_1 + 1, \dots, k_2\} \times (a, b) \}.$$

Thus $\mathcal{F}_{\text{ext}}^{(R)}$ contains all curves outside the resampling window, together with the entrance and exit data at times a and b . When the rectangle is clear from context, we simply write $\mathcal{F}_{\text{ext}}$.

Definition of the Brownian Gibbs property. We say that $\mathcal{L}$ satisfies the Brownian Gibbs property if, for every such rectangle R , the conditional distribution of

$$(\mathcal{L}_{k_1}, \dots, \mathcal{L}_{k_2})|_I$$

given $\mathcal{F}_{\text{ext}}^{(R)}$ is given by Brownian bridges with the same entrance and exit data, conditioned to avoid each other and the adjacent curves.

More precisely, set $k = k_2 - k_1 + 1$ and define

$$\begin{aligned} \bar{x} &= (\mathcal{L}_{k_1}(a), \mathcal{L}_{k_1+1}(a), \dots, \mathcal{L}_{k_2}(a)), \\ \bar{y} &= (\mathcal{L}_{k_1}(b), \mathcal{L}_{k_1+1}(b), \dots, \mathcal{L}_{k_2}(b)). \end{aligned}$$

Define the upper and lower boundary curves by

$$\begin{aligned} f &= \begin{cases} \mathcal{L}_{k_1-1}, & k_1 - 1 \in J, \\ +\infty, & k_1 - 1 \notin J, \end{cases} \\ g &= \begin{cases} \mathcal{L}_{k_2+1}, & k_2 + 1 \in J, \\ -\infty, & k_2 + 1 \notin J. \end{cases} \end{aligned}$$

Let $\mathcal{B}_{k;\bar{x},\bar{y}}^I$ denote the law of k independent Brownian bridges on I , with entrance data $\bar{x}$ at time a and exit data $\bar{y}$ at time b . Define the acceptance event

$$\text{NoTouch}_{k,I}^{f,g} := \{f(t) > B_1(t) > \dots > B_k(t) > g(t) \text{ for all } t \in (a, b)\},$$

where $(B_1, \dots, B_k)$ is sampled according to $\mathcal{B}_{k;\bar{x},\bar{y}}^I$.

Then the Brownian Gibbs property says that, almost surely,

$$\begin{aligned} \mathbb{P} \left((\mathcal{L}_{k_1}, \dots, \mathcal{L}_{k_2})|_I \in \cdot \mid \mathcal{F}_{\text{ext}}^{(R)} \right) &= \mathcal{B}_{k;\bar{x},\bar{y}}^I \left(\cdot \mid \text{NoTouch}_{k,I}^{f,g} \right) \\ &:= \mathbb{P}_{\text{NI}}^{I,\bar{x},\bar{y},f,g}. \end{aligned}$$

2 Tutorial 5: Problems

2.1 Core Problems

Problem 1 (Conditional Change of Measure). Consider two probability measures $\mathbb{P}$ and $\tilde{\mathbb{P}}$, with expectations $\mathbb{E}$ and $\tilde{\mathbb{E}}$ respectively. Assume that $\tilde{\mathbb{P}}$ is absolutely continuous with respect to $\mathbb{P}$, and let

$$\mathcal{R} := \frac{d\tilde{\mathbb{P}}}{d\mathbb{P}}.$$

For a σ -algebra $\mathcal{F}$, define

$$\mathcal{P}_{\mathcal{F}}(A) := \mathbb{E}[\mathbb{1}_A \mid \mathcal{F}], \quad \tilde{\mathcal{P}}_{\mathcal{F}}(A) := \tilde{\mathbb{E}}[\mathbb{1}_A \mid \mathcal{F}].$$

Show that, for every event A ,

$$\tilde{\mathcal{P}}_{\mathcal{F}}(A) = \frac{\mathbb{E}[\mathbb{1}_A \mathcal{R} \mid \mathcal{F}]}{\mathbb{E}[\mathcal{R} \mid \mathcal{F}]}.$$

Problem 2 (Dyson Brownian Melons have the Brownian Gibbs Property). Fix $T > 0$ and let $\mathbf{x}, \mathbf{y} \in \mathbb{R}_{>}^n$. Let $W = (W_1, \dots, W_n)$ be the law of n independent Brownian bridges on $[0, T]$ from $\mathbf{x}$ to $\mathbf{y}$, conditioned on the event

$$W_1(t) > W_2(t) > \dots > W_n(t), \quad t \in [0, T].$$

Show that W satisfies the Brownian Gibbs property.

Problem 3 (Brownian Regularity from the Brownian Gibbs Property). Let $I \subset \mathbb{N}$ and let $\mathcal{L} = \{\mathcal{L}_i\}_{i \in I}$ be a line ensemble satisfying the Brownian Gibbs property on a domain $D \subset \mathbb{R}$.

Prove the following properties.

- (a) Fix $i \in I$ and a compact interval $[a, b] \subset D$ with $a < b$. Show that $\mathcal{L}_i$ has a unique maximizer on $[a, b]$ almost surely.
- (b) Show that, for every $i \in I$, the curve $\mathcal{L}_i$ is nowhere differentiable on the interior of D , almost surely.

2.2 Bonus Problem

Problem 4 (Lower Bounding Acceptance Probabilities). Fix $T \geq 1$ and let

$$\mathbf{z} = (n, n-1, \dots, 1).$$

Let

$$B \sim \mathbb{P}_{\text{NI}}^{[-T, T], \mathbf{z}, \mathbf{z}}, \quad \tilde{B} \sim \mathbb{P}_{\text{free}}^{[-T, T], \mathbf{z}, \mathbf{z}}$$

be, respectively, n non-intersecting Brownian bridges and n independent Brownian bridges on $[-T, T]$, both starting and ending at $\mathbf{z}$. Show that there exists a universal constant $C > 0$ such that, for any event E in the path space,

$$\mathbb{P}_{\text{NI}}^{[-T, T], \mathbf{z}, \mathbf{z}}(E) \leq e^{CnT} \mathbb{P}_{\text{free}}^{[-T, T], \mathbf{z}, \mathbf{z}}(E).$$

You may use the following estimate from Tutorials 1 and 2: if B is a standard Brownian motion (or bridge) on $[0, T]$ with $B(0) = 0$ (and $B(T) = 0$ in the bridge case), then for all $\epsilon > 0$ small enough,

$$\mathbb{P} \left(\sup_{x \in [0, T]} |B(x)| \leq \epsilon \right) \geq c \exp \left(\frac{dT \log \epsilon}{\epsilon^2} \right).$$